

10

Summary and Final Remarks

10.1 SUMMARY

As of this writing, the author is not quite sure what the title of this book will be. Many well-meaning friends have suggested that it be short and snappy (and rightly so). The problem is of course that so many seemingly diverse subjects have been investigated. It is hoped that the reader will realize that there is a common thread holding them all together. To facilitate this process, the following comprehensive summary is presented next.

10.1.1 Broadband Arrays

Of all the subjects considered in this book, none has turned out to be more controversial than the broadband array treated in Chapter 6. Actually, it started quite innocently. Two of my old friends from the Harris Corporation, Bill Crosswell and Bob Taylor, stopped by my office in the OSU ElectroScience Laboratory and said, “Ben, we need a broadband array. Our ideas so far have not panned out. Do you have any suggestions?”

In fact, the fundamental principles for broadband arrays had been known not only to the author but also to several of his students for a good many years. I think it was Kornbau who in his dissertation first put in writing that a closely spaced array of dipoles without a ground plane can be designed to be very broadband (this observation was a by-product of our investigation of the Gangbuster element discussed in reference 91).

The effect of a ground plane was also well understood, but no particular interest was ever visible in the technical community. In fact, when the author mentioned the concept to a group of engineers, he was politely laughed out of the room with comments like “We do not need an overpopulated array!” Which of course would drive the price tag up. And nothing can make tears run down faster over the fat cheeks of managers than a frontal attack on their billfold.

For years there was (and in some camps still is) the unrealistic hope that grating lobes could be dealt with in some magic fashion for example by “rattling around” the interelement spacings (that may help in some scan planes but will sure make it worse in others).

Today it is probably prudent to state that there is no real alternative to the fact that the element spacings in a quality broadband array should not exceed $\sim 0.6\lambda$ at the highest frequency (for moderate scan). And that again demands hardware such as amplifiers and phase shifters that are small in size. The author is aware and extremely pleased to know that strong efforts in that respect is underway at several places in the industry. If at the same time we could design the amplifiers with balanced input terminals to be connected directly to the balanced dipoles, a classical problem would be overcome, namely designing the baluns. Although these can be made very broadband indeed (see, for example, reference 116), it is expected that a balanced amplifier can do an even better job.

It cannot be emphasized enough that the success of designing a broadband array depends largely on how well the reactive groundplane impedance is being compensated by the reactive part of the antenna impedance. At this point in time it appears that not much can be done to increase the groundplane reactance for example by using materials with higher intrinsic impedance than that of free space. If that was possible, it could lead to larger bandwidth.

Thus, the groundplane impedance will typically remain infinite at the center frequency and inductive and capacitive at the lower and higher frequencies, respectively. Consequently, only certain types of elements are suitable for broadband arrays. The flat spiral, for example, is not a good choice because its impedance changes repeatedly as a function of frequency between the inductive and capacitive region (this is typically for all long wire antennas like the two arms in a flat spiral).

Note in particular that we are not interested in an array that without a groundplane has a constant impedance over a broad band. On the contrary, we need an array impedance with a reactive part that can at least partly cancel the groundplane reactances at the upper and lower frequencies, respectively.

An array of dipoles with a groundplane can typically be designed to have a bandwidth of around 4:1 (with $VSWR < 2$). Further increase in bandwidth can be obtained by placing one or more dielectric slabs directly in front of the dipole elements as shown schematically in the equivalent circuit shown in Fig. 10.1.

This increase of bandwidth is explained (not calculated) in the following way:

1. The impedance $2R_{A0}$ in free space to the left is transformed into a lower value Z_{1-} via the dielectric slab with intrinsic impedance $2R_{A1} < 2R_{A0}$.

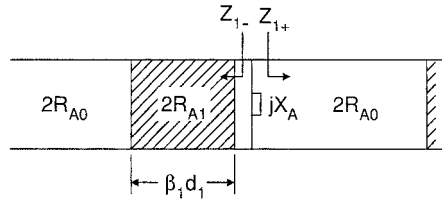


Fig. 10.1 The equivalent circuit for an array of dipoles backed by a groundplane (right) and with a dielectric slab of thickness d_1 in front. For further details see Chapter 6.

The groundplane reactance Z_{1+} is obviously not affected by the dielectric. Thus, we have in effect a lower array impedance in parallel with the same groundplane reactance which is equivalent to a larger bandwidth.

2. Furthermore, when transforming $2R_{A0}$ through the dielectric slab, we will observe a capacitive component of Z_{1-} at the lower frequencies (try it out in the complex plane and you will see it). This component will be added to jX_A in parallel, and they will both “fight” the groundplane reactance that normally has the “upper hand.”

In addition, it is well known that a dielectric slab reduces the variation of the scan impedance with scan angle (see reference 117).

This type of array has a bandwidth of about 9:1. However, optimization could increase it further. It is covered by US patent no. 6512487

10.1.2 On Antenna RCS and Edge Effect

For a number of years the author taught a graduate antenna course at Ohio State University. In the very first lecture he would usually remind the students that there are three important issues in an antenna’s life:

1. The impedance
2. The radiation pattern (from where we can obtain the directivity by integration).
3. And finally the antenna radar cross section (RCS).

The first two items got plenty of attention. However, not much time could be devoted to the antenna RCS, which was a shame since the RCS of some antennas can potentially be larger than that of an airplane.

In Chapter 2 we presented the theoretical background for antenna scattering in general. However, typically our examples were primarily aimed toward antennas with low RCS. (The demand for antennas with high RCS appears to be rather limited! There are plenty of them out there anyway.) Arrays were of particular interest by the fact that their residual scattering is basically equal to zero. More precisely, this means that if an array is conjugate-matched, not only will we

receive maximum energy (true for all antennas) but the RCS of the array will at the same time also be zero (unique for arrays).

Furthermore, in Chapter 5 we expanded our investigation to also include edge effects and showed that zero RCS is still possible. It was done rigorously by calculating the bistatic scattered field from each triad composed of a column of active elements and two columns of passive elements acting as reflectors. When conjugate-matched, the backscatter from such a triad was equal to zero.

In addition, we also found the backscattered field at other frequencies where we no longer had conjugate match. We found that the reflected field was proportional to the magnitude of the reflection coefficient for the signal incident upon the array. Furthermore, the magnitude of the reflection coefficient observed at the terminals looking out was practically identical to the one observed from the outside. This is further proof that the residual scattering in the backscatter direction for even a finite array is basically zero (normal angle of incidence).

Finite arrays will often exhibit strong variation of the scan impedance from element to element. This phenomenon is often referred to as *jitter*. This makes it difficult to obtain precise matching, which is important to obtain a low RCS (see Chapters 2 and 5). In the latter chapter we show that by connecting each element to an amplifier (conjugate-matched), we obtain a strong attenuation of any potential type II surface wave over a significant frequency band (see also Section 5.7). An additional edge treatment is sometimes recommended for perfection. Knowing that it is possible to design arrays free of jitter is extremely important.

So far we have considered planar arrays with a pencil beam. However, there is also interest in antennas with omnidirectional pattern and low RCS. Intuitively, many designers feel that such an antenna is inconsistent with the fundamentals of physics. Usually this conclusion is based on the facts associated with a simple dipole or monopole. However, we show in Chapter 7 that if we instead consider a circular array of dipoles arranged around a conducting cylinder, we can indeed obtain an omnidirectional pattern and very low RCS for signals incident anywhere in the equatorial plane. A substantial reduction was obtained over a frequency range of one octave.

It should be noted that these results were obtained with merely lumped load resistors connected directly to the dipole terminals without any matching network to improve the bandwidth as discussed in Appendix B. Application of that technique is expected to increase the bandwidth further. Further improvement is expected if dielectric sheets are placed around the dipoles analogous to the dielectric slabs in front of a planar array.

Parabolic reflector antennas cannot compete with arrays in respect to low RCS. However, by using a newly designed feed, it can be reduced substantially. Since this type of antenna still is popular in some camps, it is discussed in Chapter 8.

10.1.3 Surface Waves: Types I and II

Nothing strikes more fear in some people's minds than dielectric slabs placed next to a periodic surface. Are we not inviting one of these terrible surface waves

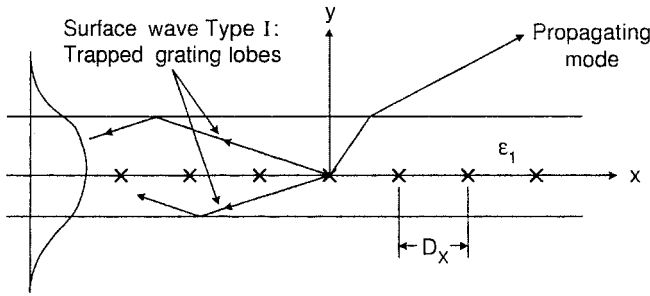


Fig. 10.2 An array of dipoles located in a dielectric slab can support a surface wave of Type I if the frequency is sufficiently high to form grating lobes inside the dielectric slab. It can exist on finite as well as infinite periodic structures.

right into our lives? We are indeed. However, once we learn to properly design periodic structures of the active as well as passive types, we can avoid many of these potential disasters.

First of all, there are basically at least two kinds of surface waves associated with periodic structures. The first kind can exist only on periodic surfaces in conjunction with a stratified medium regardless of whether the structure is infinite or finite. Thus, this type will readily be encountered when using computer codes based on infinite array theory such as the PMM program. It has therefore been known for some time and is discussed in detail in reference 118. We shall refer to this kind as Type I. As illustrated in Fig. 10.2, this surface wave consists merely of a pair of grating lobes trapped inside the dielectric slab. Only the propagating mode is incident upon the dielectric interface from within at an angle lower than the critical and will therefore make it out into free space. The trapped grating lobe will occur only at the higher frequencies where D_x/λ_1 is sufficiently large to launch grating lobes. Associated with this type of surface wave will be a field that is sinusoidal inside the slab and evanescent outside as shown in Fig. 10.2. If the structure including the dielectric slab is finite, this field will radiate in the X direction.

The second kind of surface wave denoted Type II can exist only on finite periodic structures with or without dielectric slabs. It will therefore not show up in computer codes based on infinite array theory such as the PMM program. But it does show up in the SPLAT code that incidentally can model only dielectric shells around the elements and not dielectric slabs. This kind has been investigated in this book but is less documented than Type I. In contrast to Type I, Type II can exist only for $D_x/\lambda < 0.5$ and if we also typically are 20–30% below resonance (for simple straight elements).

The important question now is, Which of the two types of surface waves is most likely to cause trouble? Well, whether we are talking about broadband arrays or FSS's the interelement spacings D_x should not exceed 0.5λ in order to obtain a good design. In other words, onset of the Type I surface wave will in general not occur before we are well above the normal operating frequency

band. However, the Type II surface wave will as stated earlier occur if $D_x < 0.5\lambda$ and if the frequency typically is 20–30% below resonance. In other words, there is a strong likelihood, if not certainty, that the Type II surface wave will be excited in a finite array operating over a broad band. If this is the case, it must be dealt with, and the treatment depends on whether the structure is active or passive.

If we are working with an active array, the good news is that in the future each element will most likely be connected to its own amplifier or generator having impedances with substantial resistive components. That is one of the most effective ways to eradicate potential Type II surface waves. An additional light edge treatment could be useful.

However, in cases where we are working with FSS's, the element loads, if there are any, will in general be entirely reactive; that is, no attenuation of a potential surface wave will take place. Nor is it a good idea to place even small resistors in each element since that would lead to reflection and transmission loss of the principal mode. In that case only a small number of edge columns should be resistively loaded. This is not as effective a way to control surface waves as resistors placed in each element, but then again FSS's are more forgiving in that respect than are the active arrays.

We finally note that curving an FSS will greatly reduce the presence of surface waves by shredding energy. Since radomes and subreflectors for example typically are nonplanar, this is probably one of the reasons that it has not been noted very much in practical applications.

10.1.4 On Broadband Matching (Appendix B)

Most engineers are quite familiar with how to design a matching network that will bring an arbitrary impedance into the center of a Smith chart at a single frequency. However, if the impedance varies with frequency (and in general it will), and they still have to place all the impedance points close to the center of the Smith chart, it will soon become clear that the conventional matching tools such as the single, the double, and even the triple stub tuner in general do a very poor job: You will lose bandwidth.

Since precision matching is an important tool for controlling the RCS of antennas in general and arrays in particular, it becomes important to be able to match the antenna over a wide band, see Chapters 2 and 5 as well as Section 10.1.2 for details. In fact, many of the author's friends have strongly suggested that this subject should be treated in a regular chapter rather than an appendix (Appendix B). My decision was guided by the fact that the subject is by no means new.¹

Thus, Appendix B should be considered more an overview of principles rather than going into every detail. You will find this in reference 119 and you might also find Appendix A useful to better understand the "tricks."

¹ It appears that the subject grew out of an effort to produce broadband antennas for countermeasure purposes during World War II. The best, if not the first, treatment appears to be given in *Very High-Frequency Techniques*. If you find a copy, grab it! [119].

10.1.5 A Broadband Meanderline Polarizer (Appendix C)

Every so often it is desirable to transform linear polarization into circular or vice versa. A meanderline polarizer is an excellent choice for such a task; and since it is a periodic structure, we have investigated it in Appendix C.

When a well-designed polarizer is exposed to an incident linear polarized field, we should first of all expect to obtain a circular polarized field of great purity. Furthermore, we should maintain a high quality over a broad band. But the real challenge is to design a polarizer where these qualities exist over a wide range of angles of incidence.

The last feature can be controlled by use of dielectric slabs on each side of the polarizer, similar to what was done earlier for scan compensation [117]. The large bandwidth (approximately an octave) was the result of careful calculation of the two orthogonal polarizations in amplitude as well as phase. It was finally finished off by a parametric study leading to a design with a bandwidth approaching an octave and angles of incidence ranging from normal up to 60° .

10.1.6 Aperiodic and Multiresonant Structures (Chapter 9)

The effect of variation of the resonant frequency of the individual elements as well as the interelement spacings was also investigated. In general, this resulted in resonance curves with extra nulls. Thus, unless such a feature is desirable for whatever reason, such a treatment is in general not recommended. However, if the structure is finite $\times$ finite instead of infinite $\times$ infinite, the nulls would disappear, leading to interesting applications.

Multiband FSS were also investigated. The complexity of such surfaces depends to a very high degree upon the frequency ratio between the desired bands.

10.1.7 The Theory

Periodic structures of finite extent are often analyzed by dividing them into infinitely long column arrays (stick arrays) with either longitudinal or transverse elements. Thus, in Chapter 3 we examined both of these types of arrays in some detail.

In the *longitudinal* case we determined the fields, being parallel as well as orthogonal to the elements. Furthermore, we obtained the mutual impedance in the spectral domain between a stick array with longitudinal elements and an outer longitudinal element as an infinite series containing products of Hankel functions and two pattern functions. Similarly, the mutual impedance between a longitudinal array and an outer orthogonal element could be obtained in the spectral domain by numerical integration of the orthogonal field along the outer orthogonal element. The orthogonal field could be expressed in the spectral domain by an infinite series containing products of Hankel functions and a single pattern function.

The beauty of the Hankel functions is that their argument $\beta r_\rho \rho$ becomes negative imaginary when $r_\rho = \sqrt{(s_z + n \frac{\lambda}{D_z})^2 - 1}$ does. In that case the Hankel

functions will fall off faster than an exponential as n increases (see Fig. 3.7). Thus, fast convergence of the infinite series is assured. This feature is quite analogous to the double infinite case where r_y goes negative imaginary, forcing the exponential term $e^{-j\beta r_y r}$ to fall off extremely fast.

The *transverse* case was somewhat more complicated as we would expect. For the field parallel with the elements, we found that it could be expressed in the spectral domain by an infinite series with terms comprised of three Hankel functions with arguments $\beta r_\rho \rho_\pm$ and $\beta r_\rho \rho_0$. Here $\rho_\pm$ and ρ_0 denotes the distance from the point of observation to the two endpoints and the midpoint of the array reference elements, respectively. The mutual impedance between the array and an outer parallel element is then obtained by numerical integration of the field along the outer parallel element.

Again as in the longitudinal case above, the argument of the Hankel function would go negative imaginary with r_ρ ensuring fast convergence. When the outer element is orthogonal to the transverse array elements, it appears at this point in time that two numerical integrations are necessary. First the orthogonal field from the array is determined by numerical integration along the array element in an infinite series containing Hankel functions. The mutual impedance is then obtained by numerical integration of the field along the outer orthogonal element. Again, the Hankel functions should insure fast convergence.

Due to these problems for the transverse array in the spectral domain, Usoff decided to write his program SPLAT primarily in the spatial domain as part of his doctoral dissertation [24]. It is extremely fast, accurate, and versatile. It simply became and currently is the workhouse for our research of finite periodic structures. It has over the years been extended and improved by several in the author's group; in particular, Dan Janning made it more user-friendly among other things.

Still, if anyone out there ever decides to try his or her hand in the spectral domain, the author would very much like to hear about it.

10.2 ARE WE GOING IN THE RIGHT DIRECTION?

Very few readers of a book pay much attention to the Preface. However, those who did may recall that I struck a somewhat critical chord concerning numerical approaches applied without appreciation of the physics. Nothing would have pleased me more than to be proven wrong. However, in the face of de facto observations getting within my range as of late, I am saddened by the fact that my pessimism was justified.

One paper in particular was investigating a combination of circuit analog and Jaumann absorbers. It was the result of a computer optimization process apparently without any consideration of physics. In less than 10 minutes I spotted the first fundamental and disastrous design flaw: The interelement spacings were larger than a full wavelength! For the author such an observation is an extremely emotional affair. It simply means that grating lobes are present in some bistatic

directions. And for some angles of incidence at some frequencies, these directions may coincide with the backscatter directions where they may produce a significant return signal. In other words, the absorber can in some cases become just the opposite: an RCS enhancer.

Another design had placed a circuit analog sheet less than 0.01λ in front of the groundplane. Even an inexperienced designer should know that a groundplane that close to a CA sheet basically shorts it out, leaving it virtually ineffective (try it out in a Smith chart and you will soon understand).

Sure, unconventional designs can sometimes lead to new superior designs. Not this time. The bandwidth was typically 25% of what should be expected from a design that complex (read Chapter 9 in reference 120).

In another paper the RCS of an array of conjugate-matched slots were calculated. I deliberately did not say investigated because only a minimum of physical insight was displayed. It should therefore surprise no one that the resulting RCS was 6 dB higher than that of a flat plate with an area equal to the total aperture! It should have been “infinitely” many decibels below (see Chapter 2). A simple sign error? Yes, but designers with physical insight would catch such a mistake almost instantly.

While the list of indiscretions goes on, let us stop here and just ask the simple question, How can researchers of even great repute ever fall into traps like these? And furthermore, How is it possible that papers of such quality can pass through the review process associated with the world's most prestigious journals?

Let us try to answer the second question first.

There are basically two reasons for anyone to undertake a review:

1. You are genuinely interested in reading about the latest in your field of interest. If you feel that you are inadequate, you simply pass the paper on to a more competent colleague. This is the proper procedure for reviewing a paper.
2. You are merely doing it because you are asked and it is expected of you in order to tip the promotion scale to the next level in the academic world. You may even secretly enjoy the power you hold over another person's reputation. Chances are you are really not well-suited to review the paper. You are reviewing it for the wrong reasons.

There is no unique answer to the first question. However, a common thread seems to be that the designer has put an almost blind trust into mindless computations only. There is often a single-minded belief that the computer will eventually find the optimum solution based entirely on its superior speed. No physical insight is necessary, or so it is thought.

However, as already shown above by several examples, this is far from the case. More specifically: If Boeing starts designing airplanes without involving their vast practical experience and know-how, my flying days will come to an end!

We should of course have an open mind for those who want to experiment with new ideas in their own backyard. But please do not publish these designs

unless they are sound. Most people expect papers written in accredited journals to be not only correct but also superior in quality. Again, the review process is often hampered because many reviewers have a minimum of engineering knowledge, only the mathematical insight. Later it then becomes almost impossible for a straight-thinking competent designer to convince the public otherwise.

And please do not try to extract money from those with limited technical knowledge that nevertheless control vast pools of money. Many of them are easily taken in by imaginative notions and fairy tales. Note that the number of publications may not mean much. As we have seen, they may be right, wrong, significant, or so what. Unfortunately it often appears that the further “out” a proposal, the more some potential sponsors might like them.

The conclusion of our discussion above is that a numerical optimization process must be treated very carefully. If the end product has to be manufactured or is important in any other way (not just for writing a paper!), it is strongly recommended to work side by side with a designer well-versed in the physics of the problem, which brings us right to the center of my deepest concern: We are not educating enough young engineers with solid knowledge of fundamentals. It sounds almost unbelievable that some professors advise their students not to take any classes where real engineering is taught (heaven forbid that the students should get smarter than their professors!).

10.3 LET US MAKE UP!

The author has several times expressed concern over using purely numerical approaches without sufficient physical insight.

But it would be unfair and biased to finish this chapter without applauding some of the remarkable results that can be obtained only by a numerical approach (short of measurements, of course). Particularly appreciated by the author is the investigation of the so-called *feed organizer* shown in Fig. 10.3.² The purpose of this device is twofold: It provides a precise and robust way of feeding two sets of orthogonal dipoles. And at the same time it appears to prevent excitation of the so-called push–push or common mode. In that case, in addition to the desired feed current to the dipoles (push–pull mode), we observe a mode where a pair of feed lines is excited like monopoles orthogonal to the groundplane and where the dipoles act simply as top-loading. If that is the case, an undesirably high sidelobe level will occur. And the terminal impedance will also be adversely affected. A detailed discussion is given, for example, in the dissertation by S.J. Lin [121].

The phenomenon is often blamed on faulty baluns. This may or may not be the case. In fact, even with perfect baluns the phenomenon can occur. What happened is simply that either the incident field or the field from the array itself will at oblique scanning have a field component along the feed lines that may

² This ingenious device was invented by Robert Taylor from the Harris Corporation, US Patent no. 6483464.

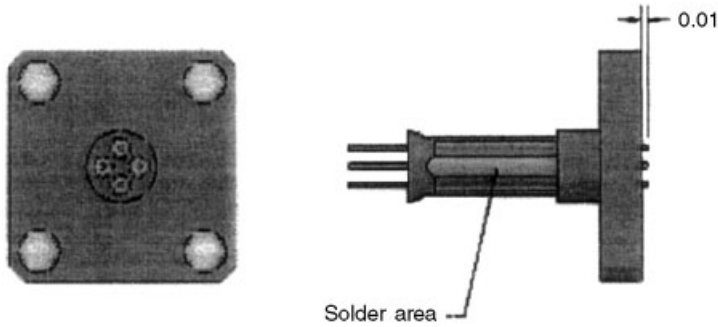


Fig. 10.3 The feed organizer comprised of a step with four holes to accommodate four coaxial cables for feeding the two sets of orthogonal dipoles. Originally intended to merely provide mechanical support, it apparently also suppresses the common or the push–push mode.

readily excite the common mode if conditions are right. What happens when we use a feed organizer is apparently that the casing of the feed organizer almost becomes part of the ground plane and consequently remains “cool.”

A detailed examination of these conditions are obtained by one of several finite element method programs like for example the HFSS code. Details of this examination will be given in forthcoming reports by Harris Melbourne, Florida. It shows the finite element method in its finest hour. Bravo!

Finally we show in Fig. 10.4 a practical execution of an array comprised of two orthogonal sets of dipoles fed at the points of their intersections. We also clearly see the finger capacitors. These should be sufficiently large to ensure proper performance at the lower frequencies. However, the length of the fingers

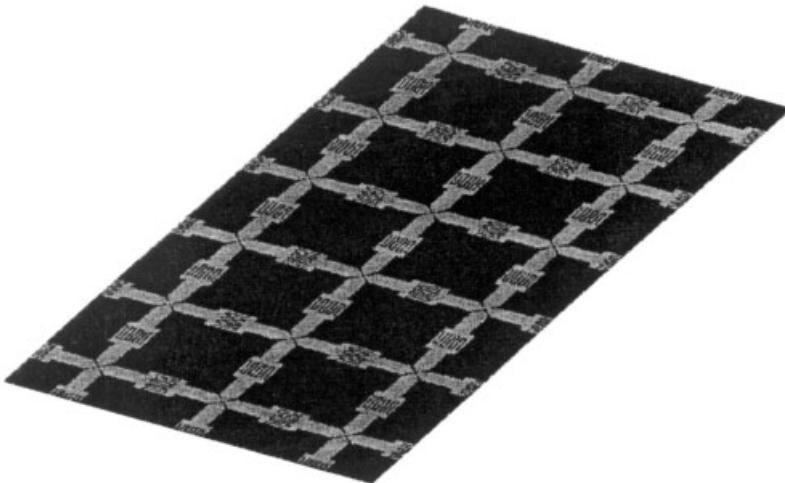


Fig. 10.4 Array of two sets of orthogonal dipoles fed at their crosspoints. Note also the finger capacitors between the feedpoints.

should not be too long since it might lead to undesirable resonances at the higher frequencies. The example shown here is made by the Harris Corporation. Measurements showed excellent agreement with the calculated curves. See also the back cover of this book.

Mission Research Corporation Dayton, Ohio has also built broadband arrays based on the principle presented in Chapter 6.